DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

"Master Data Checker Tool"

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

BHAVITH P B RAO - 1DS20AI017

Conducted at

Alcon Laboratories Pvt. Ltd

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangelore-78

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

CERTIFICATE

This is to certify that the Internship titled **“Master Data Checker Tool”** carried out by **Mr. BHAVITH P B RAO[1DS20AI017]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (***20AI8ICINT***)

Sajjad, Tuhin

Digitally signed by Sajjad, Tuhin
DN: cn=Sajjad, Tuhin, ou=PDF Internal Signing, o=Alcon Vision, LLC
Reason: I am approving this document
Date: 2024.05.09 16:13:50 +05'30'

Signature of Reporting Head	**Signature of Internship Coordinator**	**Signature of HoD**
Tuhin Sajjad	**Prof. Kavya D N**	**Dr. Vindhya P Malagi**
Senior Manager, MTOQA Platforms Business Technology Alcon Laboratories	Dept. of AI & ML DSCE	Dept. of AI & ML DSCE

External Viva:

Name of the Examiner	Signature with Date
1) __________________	________________
2) __________________	_________________

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

D E C L A R A T I O N

I, **BHAVITH P B RAO,** final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Alcon Laboratories Pvt. Ltd**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date: USN: 1DS20AI017

Place: Bangalore NAME: BHAVITH P B RAO

OFFER LETTER

Alcon
SEE BRILLIANTLY

Date: 22 January 2024

Bhavith P B Rao
Dayananda Sagar College of Engineering
Shavige Malleswaram Hills, 91st Main Rd,
1st Stage, Kumaraswamy Layout,
Bengaluru, Karnataka 560078

AGREEMENT

SUB: Internship with Alcon Laboratories (India) Pvt. Ltd.

We would like to congratulate you on being selected for an internship with **Alcon** based at **Bangalore**.

Your internship is scheduled to start effective **5 Feb, 2024** for a period of **6 months** as per the internship request letter issued by **Dayananda Sagar College of Engineering** with letter dated **4 Dec 2023**.

You should report for the internship at the following address:

Alcon Laboratories (India) Pvt. Ltd.
Godrej Woodsman Estate RMZ Azure (IT Complex),
9th, 10th & 11th, Bellary Rd, Hebbal,
Bengaluru, Karnataka 560092

Reporting Time: 10:00 AM
Contact Person: Santhosh Kumar SG
Department: Human Resources (Talent Acquisition)

You will be eligible to receive a **stipend** of **INR 25,000 per month** during your internship. Other terms are mentioned in Annexure enclosed with this letter.

We look forward to welcome you and wishing you an enriching learning experience with us.

Sincerely,
Indumathy Karthikeyan
HR Head – AGS Bangalore, India

WE ARE
GREAT PLACE TO WORK - CERTIFIED™

Acknowledgement:

I hereby acknowledge, confirm, and agree to the contents of this letter including the Annexure.

Bhavith P B Rao

Annexure

1. The internship or training is in accordance with the internship letter requested by **Dayananda Sagar College of Engineering** with letter dated **Dec 4, 2023,** that he/she has qualified all prerequisites including examination or other requirements to undergo the internship or training.
2. Intern shall, faithfully and to the best of their ability, devote all working hours exclusively to the performance of the learning as may be assigned by Alcon from time to time. Intern shall abide by applicable laws, guidelines, and policies of Alcon during internship.
3. During internship, Intern will have access to business, confidential and personal information ('Confidential Information') of the Company, its associates, customers, vendors, or other parties. Intern shall at any time during or after the internship maintain confidentiality of the information and shall not make any unauthorized disclosure of Confidential Information or make any use of Confidential Information except for purposes permitted and intended by the Company.
4. Either party is entitled to terminate the internship with 30 days prior notice Employment and Conflicts of Interest

COMPLETION CERTIFICATE

A C K N O W L E D G E M E N T

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mr. Tuhin Sajjad, Alcon Laboratories, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

BHAVITH P B RAO
[1DS20AI017]

TABLE OF CONTENTS

CHAPTER 1
INTRODUCTION

Introduction to Master Data Checker Tool

A meticulously designed and developed tool addresses the challenges of validating master data before transfer processes, offering three key use cases. Firstly, it ensures the presence of all dependencies within designated folders before exporting the transfer package, identifying any missing dependencies as warnings. Additionally, it also checks for the existence of master data objects and their dependencies in the target system prior to import, reporting any missing data to prevent issues during import. Lastly, it verifies the accuracy and completeness of transferred data by comparing master data in both environments, flagging any missing data for prompt remediation, thus ensuring data integrity and consistency across environments.

1. LIMS

In order to manage various aspects at Alcon, a robust and versatile management software is required. A Laboratory Information Management System (LIMS) is a software solution designed to manage and automate the workflows and data management processes within a laboratory environment. LIMS is used across various industries, including pharmaceuticals, healthcare, environmental testing and more. Key features of LIMS include Sample Management, Data Management, Workflow Automation, Instrument Integration, Quality Control, Inventory Management, Reporting and Analytics, Audit Trails and Security, etc.

2. Master Data

Master Data forms the cornerstone of most information systems, serving as the static data that establishes the standard for critical business information. Within the Laboratory Information Management System (LIMS), master data encompasses vital components such as Analyses, Batches, Instrumentation, and Samples, among others. As such, the implementation processes surrounding Master Data profoundly influence the maintenance and functionality of systems like LabVantage LIMS throughout their operational lifespan. Master data management involves various tables, each containing multiple custom fields, each serving specific functionalities within the system. Given the sensitive nature of industries utilizing LIMS, such as health and pharmaceuticals, meticulous management of master data is paramount to ensure regulatory compliance and operational efficiency.

3. GxP Environment:

GxP environments refer to Good Practice environments, which are environments that follow good practices in areas such as manufacturing, distribution, and quality control. These practices are often required in industries such as pharmaceuticals and food production to ensure the safety and quality of products. The term GxP is derived from acronyms such as GMP (Good Manufacturing Practice), GDP (Good Distribution Practice), and GCP (Good Clinical Practice).

4. Data Migration

In the realm of LIMS, master data management typically unfolds across three environments: development, validation, and production. The validation and production environments adhere to GxP standards, encompassing quality guidelines and regulations for good practices. Initially, master data is created by laboratory managers, Subject Matter Experts and R&D personnel within the development environment. To transition this data into the GxP environments, rigorous testing and validation procedures are imperative to verify its accuracy and integrity. Upon successful validation, error-free data is transferred from the development environment to the validation environment, where documentation of validations occurs. Subsequently, the validated data is migrated to the production environment for operational use, ensuring seamless functionality and compliance within the LIMS framework.

CHAPTER 2
DESCRIPTION OF ORGANIZATION

ALCON LABORATORIES

Alcon
SEE BRILLIANTLY

Introduction

Alcon is the global leader in eye care, dedicated to helping people see brilliantly. With an over 75-year heritage, it is the largest eye care device company in the world, with complementary businesses in Surgical and Vision Care. It operates in the ophthalmic surgical and vision care markets, which are large, dynamic and growing. Being a truly global company, they have their presence in over 50 countries and serve patients in more than 140 countries. It has a long history of industry firsts, and commits a substantial amount in Research and Development to meet customer needs and patient demands.

Know more about them on: Developing Innovative Eye Care Treatments | Alcon.com

ABOUT THE COMPANY

About Alcon Laboratories

Alcon Inc. is a Swiss-American pharmaceutical and medical device company specializing in eye care products. It has a paper headquarters in Geneva, Switzerland but its operational headquarters are in Fort Worth, Texas, United States, where it employs about 4,500 people. Alcon was a subsidiary of Novartis until April 2019, when it was spun out into a separate publicly traded company. Alcon itself has a number of subsidiaries, including Aerie Pharmaceuticals, focused on therapies for glaucoma and other diseases of the eye, and WaveLight, which develops and manufactures laser eye surgery technologies.

VISION STATEMENT

We help people see brilliantly.

MISSION STATEMENT

At Alcon, we are laser-focused on creating the exceptional experience - not just for our customers, but for our associates, too. Achieve more together.

CHAPTER 3
SCOPE OF WORK

PROBLEM STATEMENT

The existing Master Data management process within the LIMS is recognized as a significant bottleneck, characterized by frequent occurrences of data integrity issues, version compatibility challenges, and configuration disparities across environments. These issues result in notable delays in data availability, heightened workload for rectifying data errors, and an elevated risk of non-compliance. The project will leverage Lean Six Sigma methodologies to pinpoint waste and inefficiencies within the current master data management process, subsequently implementing process enhancements. This includes the development of a tool for pre-validating dependent, missing, or inaccurate data, along with automated validation tools for assessing data integrity post master data promotion in the Target environment. Additionally, documentation and evidence will be provided for comparing data before and after promotion in both the Target and Source environments.

CHAPTER 4
LEARNING OBJECTIVES

1. Mastery of Java and SQL Basics:

Interns are required to go through online courses to gain comprehensive proficiency in Java programming language and SQL fundamentals. Mastery of Java involves understanding the core concepts of object-oriented programming (OOP), including classes, objects, inheritance, polymorphism, and encapsulation, as well as advanced topics like multithreading, exception handling, and generics. Additionally, proficiency in SQL basics involves grasping essential concepts such as query statements, understanding RDBMS principles, database design, normalization, indexing, and basic transaction management. Achieving mastery in both Java and SQL basics enables individuals to develop robust software applications, interact with databases efficiently, and effectively manage data.

2. LIMS and Product Knowledge:

Interns undergo training on LIMS and product knowledge relevant to Alcon. LIMS is software designed to manage and track samples, data, and workflows in laboratory settings, offering functionalities such as sample tracking, data analysis, instrument integration, and compliance management. Understanding LIMS involves familiarity with its architecture, functionalities, configuration, and customization capabilities to meet specific laboratory requirements. Additionally, acquiring product knowledge within the healthcare industry involves understanding the features, functionalities, and applications of products or services offered within that industry. This knowledge encompasses understanding the market landscape, competitor analysis, customer needs, product lifecycle management, and regulatory compliance, enabling individuals to effectively market, sell, support, or develop products tailored to industry requirements.

3. **Alcon Standard Practices and Ideologies:**

Interns focuses on understanding and adhering to the standardized procedures, best practices, and core ideologies upheld by Alcon. This entails gaining insight into Alcon's established methodologies, protocols, and guidelines across various functions. It also involves embracing Alcon's core values, mission, vision, and corporate culture, which emphasize innovation, excellence, integrity, collaboration, and customer-centricity. By mastering these, individuals can align their work processes, decision-making, and behavior with the company's objectives, fostering consistency, efficiency, and alignment with organizational goal.

CHAPTER 5
CHALLENGES AND SOULTION

The Master Data management process within the LIMS team has been plagued by several critical issues, manifesting as major bottlenecks within the system:

1. **Data Integrity Issues:** Frequent lapses in data integrity compromise the reliability and accuracy of the Master Data stored within the LIMS. These discrepancies undermine the trustworthiness of the data and impede crucial decision-making processes within the organization.
2. **Version Compatibility Problems:** Incompatibility between different versions of Master Data creates interoperability challenges and hampers the seamless exchange of information across various modules and environments. This disparity leads to inefficiencies in data management and impedes the timely execution of critical tasks.
3. **Configuration Discrepancies:** Discrepancies in the configuration settings of the Master Data within the LIMS exacerbate operational challenges and introduce inconsistencies across different environments. This lack of uniformity in configurations complicates data management processes and increases the likelihood of errors and compliance failures.
4. **Delays in Data Availability:** The aforementioned issues contribute to significant delays in the availability of Master Data for use in operational activities and decision-making processes. These delays hinder the organization's responsiveness to changing business requirements, adversely impacting overall performance and productivity.
5. **Increased Workload for Data Correction**: The prevalence of data integrity issues necessitates extensive efforts and resources for data correction and reconciliation. This increased workload not only strains operational resources but also diverts valuable time and attention away from other critical tasks and initiatives.
6. **Higher Risk of Compliance Failures:** The accumulation of data integrity issues, version compatibility problems, and configuration discrepancies heightens the organization's vulnerability to compliance failures and regulatory non-compliance. These risks pose significant threats to the organization's reputation, financial stability, and regulatory standing.

Addressing these challenges is paramount to optimizing the Master Data management process within the LIMS and enhancing overall operational efficiency and compliance.

Solution:

A tool has been meticulously designed and developed to address the challenges inherent in validating the master data before the transfer processes. This tool encompasses three main use cases aimed at ensuring the integrity and completeness of master data across source and target environments: Pre-validate Export, Pre-validate Import, and Post-validate Import.

1. **Pre-validate Export:** Before exporting the transfer package from the source environment, it is imperative to ascertain that all dependencies of the master data, which are organized into folders, are present within the designated folder. This use case focuses on identifying any missing dependencies within the folder structure. If discrepancies are detected, they are reported as warnings, indicating the absence of certain dependencies in the folder.

2. **Pre-validate Import:** Prior to importing the package into the target environment, thorough checks are conducted to determine whether the master data objects, along with their dependencies, exist within the target system. This use case ensures that the necessary master data components are available in the target environment to facilitate a seamless import process. Any missing data in the target system is promptly identified and reported to mitigate potential issues during the import process.

3. **Post-validate Import:** Following the successful import of master data into the target environment, the Post-validate Import use case comes into play. This phase involves verifying the accuracy and completeness of the transferred data. A meticulous comparison is made between the master data in the source and target environments to identify any disparities. Any data that was not successfully transferred to the target environment is flagged as missing, enabling prompt remediation to ensure data integrity and consistency across environments.

The tool has been developed utilizing Java with Maven, leveraging the robustness and flexibility of these technologies to ensure efficient and scalable implementation. Java provides a versatile and platform-independent programming language, while Maven facilitates project management and dependency resolution, streamlining the development process.

To interact with the data stored in Oracle SQL Developer, SQL queries are utilized within the Java application. This allows seamless access to the master data stored in the database, enabling comprehensive validation and comparison processes.

Furthermore, the tool's functionality is encapsulated within executable JAR files, simplifying deployment and execution across different environments. Batch files are utilized to initiate the application, providing a straightforward and user-friendly interface for running the tool.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

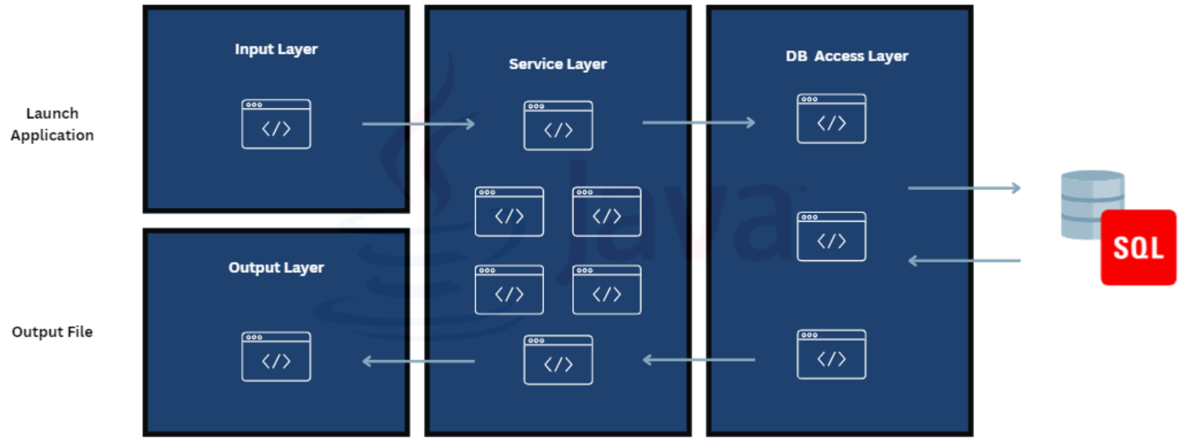
- Front End: Windows BAT File
- Data Base: Oracle SQL
- Runtime: OpenJDK 11

Hardware Requirements:

- Processor: Intel Core i5 Processor
- Memory: 8GB
- Hard Disk: 20 GB

CHAPTER 6
ACHEIVEMENTS AND CONTRIBUTION

System Design:



The Master Data Checker Tool mainly contains:

- **Input Layer** - The Input Layer of an application is responsible for handling incoming data or requests from external sources, such as users, other systems, or devices. It serves as the entry point for data into the application. This layer is often responsible for parsing and validating incoming data, converting it into a format that the application's internal components can process, and routing it to the appropriate parts of the system.
- **Service Layer** - The Service Layer contains the business logic and application-specific functionality of the system. It encapsulates the core operations and processes that the application performs to fulfill requests or manipulate data. This layer typically orchestrates interactions between different components of the application, such as data access, validation, transformation, and external integrations.
- **DB Access Layer** - The DB Access Layer is responsible for interacting with the underlying database or data storage systems. It encapsulates all database operations, such as querying, updating, inserting, and deleting data, as well as managing transactions and connections. This layer shields the rest of the application from the complexities of database interaction and provides a consistent interface for accessing and manipulating data,

regardless of the underlying database technology or schema.

- **Output Layer** - The Output Layer handles the generation and delivery of responses or output from the application back to the external entities that initiated requests .It formats and transforms data into the appropriate representations for delivery to clients, which could include rendering HTML for web browsers, generating JSON or XML for APIs, or producing files for download.

The successful implementation of the Master Data Checker Tool has yielded notable accomplishments, both quantitatively and qualitatively.

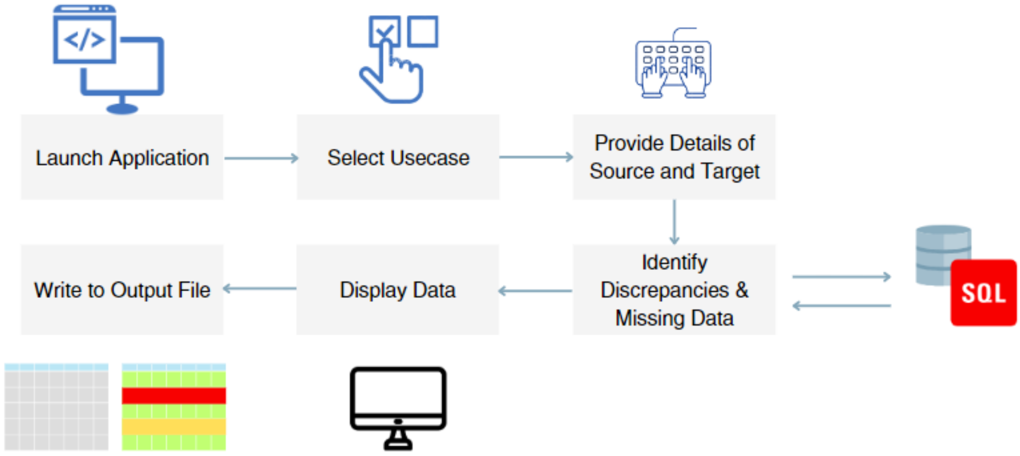
Quantitative:

- Cost Savings: Implementation of the Master Data Checker Tool resulted in a significant reduction in data-related errors, leading to fewer retests and corrections. This initiative is estimated to save approximately $50,000 annually, contributing to enhanced cost efficiency and resource optimization.
- Increased Throughput: The decreased cycle time facilitated by the tool has substantially enhanced the agility of laboratory operations. This improvement potentially increases throughput by 20%, allowing for more efficient utilization of resources and better meeting organizational demands.

Qualitative:

- Enhanced Compliance: The project has led to improved data integrity and system reliability, thereby enhancing compliance with regulatory standards. By ensuring accuracy and completeness of master data, the tool supports adherence to industry regulations and mitigates compliance risks.
- Enhanced User Satisfaction: The streamlined and less error-prone processes facilitated by the Master Data Checker Tool have resulted in better user satisfaction. With smoother workflows and reduced data discrepancies, users experience greater efficiency and confidence in system operations, contributing to overall job satisfaction and productivity.

Flow of the process



Upon initiation of the batch file, the application prompts the user to provide required information regarding the source and target databases to establish connections. Additionally, input regarding the folder and the user who created the folder, as per the specific use case requirements, is taken from the user.

Once the necessary information is obtained, the application proceeds to establish connections with the specified databases. Subsequently, the relevant data is retrieved from the respective database(s), facilitating a comprehensive analysis of the master data contained within.

The core logic of the application involves identifying dependencies associated with the master data objects and executing comparison algorithms to determine any disparities between the source and target environments. This intricate process ensures thorough validation and verification of the master data transfer, mitigating risks associated with missing dependencies.

Upon completion of the comparison process, the application generates detailed reports outlining any missing data or discrepancies encountered during the validation phase. These reports are meticulously formatted and written onto an output file, which can be configured as either a text file or an Excel spreadsheet, depending on the use case.

CHAPTER 7
REFLECTION AND ANALYSIS

My internship experience has been incredibly enriching and transformative, providing me with invaluable lessons and insights that will shape my professional journey moving forward. One key lesson I learned is the importance of proactive networking and engaging with global leaders. These interactions not only broadened my perspective but also enhanced my understanding of navigating corporate dynamics and challenges. I realized the significance of effective communication and leadership skills in fostering meaningful connections and driving collaboration across diverse teams.

Moreover, the technical skills I acquired, particularly in Java programming and SQL database management, have significantly strengthened my capabilities as a software engineer. The hands-on experience gained from working on real-world projects, such as developing the Master Data Checker tool, equipped me with practical knowledge and problem-solving skills essential for success in the industry.

In the future, I plan to apply these newly acquired skills and insights to pursue opportunities for innovation and excellence in my career. Whether it's designing robust software solutions, facilitating cross-functional collaboration, or leading initiatives to drive organizational growth, I am confident that the skills honed during my internship will be instrumental in achieving my professional goals.

While reflecting on my internship experience, I also recognize areas for further development and improvement. Specifically, I aim to enhance my project management skills and deepen my understanding of industry best practices. To address these areas, I plan to seek out relevant training and mentorship opportunities, actively engage in challenging projects, and continuously strive for personal and professional growth.

Overall, my internship at Alcon Laboratories has been a transformative journey that has equipped me with the skills, knowledge, and confidence to thrive in the dynamic and competitive landscape of the software industry. I am grateful for the opportunity and look forward to applying my learnings to make meaningful contributions in my future endeavours.

OUTPUT

Prevalidate Import

Input and execution on command prompt:

```
2024-04-29 11:43:17 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriver - Bootstrapping App
2024-04-29 11:43:17 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriver - Running App
Source DB:
1. MD
2. CFG
3. STG
4. VAL
5. PROD
6. SBX
Enter your choice:6
2024-04-29 11:43:22 [main] INFO  com.alcon.mdlims.masterdata.service.DBConnectionService - Database connection successful with ID: adc88d5c-8950-4b71-8d50-98b14db99d63
Source Schema:LVSBX87
Target DB:
1. MD
2. CFG
3. STG
4. VAL
5. PROD
6. SBX
Enter your choice:2
2024-04-29 11:43:31 [main] INFO  com.alcon.mdlims.masterdata.service.DBConnectionService - Database connection successful with ID: 7cc30e3c-228b-4575-a973-0ad5761439f9
Target Schema:LVCFG87
Method type:METHOD
Folder Name:C19
User ID:SRA3
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Usecase: Prevalidate Import
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Source Connection:oracle.jdbc.driver.T4CConnection@62e20a76
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Source Schema: LVSBX87
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Target DB: oracle.jdbc.driver.T4CConnection@1d1f7216
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Target Schema: LVCFG87
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Method Type: METHOD
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Folder Name: C19
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - User ID: SRA3
2024-04-29 11:43:50 [main] INFO  com.alcon.mdlims.masterdata.dao.HierarchyRetrieval - Query for WORKITEM: SELECT TESTINGDEPARTMENTID, SECURITYDEPARTMENT, WORKAREADEPARTMENTID, EMBEDCHILDSAMPLEPLANID, EMBEDCHILDS
AMPLEPLANVERSIONID, APPLICABLESAMPLETYPEID, SECURITYUSER, AUTOASSIGNANALYSTID, QUANTITYUNITS, WORKITEMID, WORKITEMVERSIONID FROM LVSBX87.WORKITEM MDTB, (select  SFI.LINKSDCID, SFI.LINKKEYID1, SFI.LINKKEYID2, SFI
.LINKKEYID3 from lvsbx87.sysuserfolderitem SFI INNER JOIN lvsbx87.sysuserfolder SUF ON SUF.sysuserid = SFI.sysuserid and SUF.sysuserfolderid = SFI.sysuserfolderid WHERE SUF.folderlabel = 'C19' and SUF.sysuserid
= 'SRA3' ) FDT WHERE  MDTB.WORKITEMID = FDT.LINKKEYID1 AND  MDTB.WORKITEMVERSIONID = FDT.LINKKEYID2
2024-04-29 11:44:37 [main] INFO  com.alcon.mdlims.masterdata.dao.HierarchyRetrieval - Query for PARAMLIST: SELECT ACCREDITEDADDRESSID, ACCREDITEDADDRESSTYPE, APPROVALTYPEID, SECURITYDEPARTMENT, TESTINGDEPARTMENT
ID, WORKAREADEPARTMENTID, S_INSTRUMENTMODEL, S_INSTRUMENTTYPE, S_INSTRUMENTTYPE, LIMITRULEID, LIMITRULEVERSIONID, SECURITYUSER, AUTOASSIGNANALYSTID, PARAMLISTID, PARAMLISTVERSIONID, VARIANTID FROM LVSBX87.PARAML
IST MDTB, (select  SFI.LINKSDCID, SFI.LINKKEYID1, SFI.LINKKEYID2, SFI.LINKKEYID3 from lvsbx87.sysuserfolderitem SFI INNER JOIN lvsbx87.sysuserfolder SUF ON SUF.sysuserid = SFI.sysuserid and SUF.sysuserfolderid =
 SFI.sysuserfolderid WHERE SUF.folderlabel = 'C19' and SUF.sysuserid = 'SRA3' ) FDT WHERE  MDTB.PARAMLISTID = FDT.LINKKEYID1 AND  MDTB.PARAMLISTVERSIONID = FDT.LINKKEYID2 AND  MDTB.VARIANTID = FDT.LINKKEYID3
2024-04-29 11:45:50 [main] INFO  com.alcon.mdlims.masterdata.domain.TextFileWriter - Data written to file: PrevalidateImport_C19_SRA3_20240429_114349.txt
2024-04-29 11:45:50 [main] INFO  com.alcon.mdlims.masterdata.domain.TextFileWriter - Text File Writer closed successfully!
2024-04-29 11:45:50 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriverHelper - SourceDB closed successfully!
2024-04-29 11:45:51 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriverHelper - TargetDB closed successfully!
```

Output:

PrevalidateImport_C19_SRA3_2024

File Edit View

```
---------------------------------------------

Missing Keys in WORKITEM SDC:
Primary Keys:
WORKITEMID = PCR Test C19       WORKITEMVERSIONID = 1

Foreign Keys: SECURITYUSER = RAOBH1
WHERE Primary Keys: WORKITEMID = PCR Test C19 WORKITEMVERSIONID = 1


---------------------------------------------

---------------------------------------------

Missing Keys in WORKITEMITEM SDC:
Primary Keys:
WORKITEMID = PCR Test C19       WORKITEMVERSIONID = 1   WORKITEMITEMID = 1
WORKITEMID = PCR Test C19       WORKITEMVERSIONID = 1   WORKITEMITEMID = 2

Only Primary Keys missing
WHERE Primary Keys: WORKITEMID = PCR Test C19 WORKITEMVERSIONID = 1 WORKITEMITEMID = 1


Only Primary Keys missing
WHERE Primary Keys: WORKITEMID = PCR Test C19 WORKITEMVERSIONID = 1 WORKITEMITEMID = 2


---------------------------------------------

---------------------------------------------

Missing Keys in WORKITEMINSTRUMENT SDC:
No Primary Keys are missing
No Foreign Keys are missing
---------------------------------------------

---------------------------------------------

Missing Keys in WORKITEMREAGENTTYPE SDC:
```

Ln 1, Col 1   3,186 characters

Prevalidate Export

Input and execution on command prompt:

```
2024-04-29 11:38:18 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriver - Bootstrapping App
2024-04-29 11:38:18 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriver - Running App
Source DB:
1. MD
2. CFG
3. STG
4. VAL
5. PROD
6. SBX
Enter your choice:6
2024-04-29 11:38:33 [main] INFO  com.alcon.mdlims.masterdata.service.DBConnectionService - Database connection successful with ID: 7a3d2169-dd99-4408-a2cd-53d
Source Schema:LVSBX87
Method type:METHOD
Folder Name:C19
User ID:SRA3
2024-04-29 11:38:58 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Usecase: Prevalidate Export
2024-04-29 11:38:58 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Source Connection:oracle.jdbc.driver.T4CConnection@62e20a76
2024-04-29 11:38:58 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Source Schema: LVSBX87
2024-04-29 11:38:58 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Target DB: null
2024-04-29 11:38:58 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Target Schema: null
2024-04-29 11:38:58 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Method Type: METHOD
2024-04-29 11:38:58 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Folder Name: C19
2024-04-29 11:38:58 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - User ID: SRA3
2024-04-29 11:39:02 [main] INFO  com.alcon.mdlims.masterdata.dao.HierarchyRetrieval - Query for WORKITEM: SELECT TESTINGDEPARTMENTID, SECURITYDEPARTMENT, WORK
AMPLEPLANVERSIONID, APPLICABLESAMPLETYPEID, SECURITYUSER, AUTOASSIGNANALYSTID, QUANTITYUNITS, WORKITEMID, WORKITEMVERSIONID FROM LVSBX87.WORKITEM MDTB, (selec
.LINKKEYID3 from lvsbx87.sysuserfolderitem SFI INNER JOIN lvsbx87.sysuserfolder SUF ON SUF.sysuserid = SFI.sysuserid and SUF.sysuserfolderid = SFI.sysuserfold
= 'SRA3' ) FDT WHERE  MDTB.WORKITEMID = FDT.LINKKEYID1 AND  MDTB.WORKITEMVERSIONID = FDT.LINKKEYID2
2024-04-29 11:39:04 [main] INFO  com.alcon.mdlims.masterdata.dao.HierarchyRetrieval - Query for PARAMLIST: SELECT ACCREDITEDADDRESSID, ACCREDITEDADDRESSTYPE, 
ID, WORKAREADEPARTMENTID, S_INSTRUMENTMODEL, S_INSTRUMENTTYPE, S_INSTRUMENTTYPE, LIMITRULEID, LIMITRULEVERSIONID, SECURITYUSER, AUTOASSIGNANALYSTID, PARAMLIST
IST MDTB, (select  SFI.LINKSDCID, SFI.LINKKEYID1, SFI.LINKKEYID2, SFI.LINKKEYID3 from lvsbx87.sysuserfolderitem SFI INNER JOIN lvsbx87.sysuserfolder SUF ON SU
 SFI.sysuserfolderid WHERE SUF.folderlabel = 'C19' and SUF.sysuserid = 'SRA3' ) FDT WHERE  MDTB.PARAMLISTID = FDT.LINKKEYID1 AND  MDTB.PARAMLISTVERSIONID = FD
2024-04-29 11:39:04 [main] INFO  com.alcon.mdlims.masterdata.domain.TextFileWriter - Data written to file: PrevalidateExport_C19_SRA3_20240429_113901.txt
2024-04-29 11:39:04 [main] INFO  com.alcon.mdlims.masterdata.domain.TextFileWriter - Text File Writer closed successfully!
2024-04-29 11:39:05 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriverHelper - SourceDB closed successfully!
```

Output:

```
PrevalidateExport_C19_SRA3_2024
File    Edit    View

-----------------------------------------------

Missing Keys in WORKITEM SDC:
No Foreign Keys are missing
-----------------------------------------------

-----------------------------------------------

Missing Keys in PARAMLIST SDC:

Foreign Keys: APPROVALTYPEID = MasterData
WHERE Primary Keys: PARAMLISTID = StabilityC19 PARAMLISTVERSIONID = 1 VARIANTID = 1


Foreign Keys: LIMITRULEVERSIONID = 1 LIMITRULEID = Pass / Fail
WHERE Primary Keys: PARAMLISTID = StabilityC19 PARAMLISTVERSIONID = 1 VARIANTID = 1


-----------------------------------------------
```

Postvalidate Import

Input and execution on command prompt:

```
2024-04-29 11:43:17 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriver - Bootstrapping App
2024-04-29 11:43:17 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriver - Running App
Source DB:
1. MD
2. CFG
3. STG
4. VAL
5. PROD
6. SBX
Enter your choice:6
2024-04-29 11:43:22 [main] INFO  com.alcon.mdlims.masterdata.service.DBConnectionService - Database connection successful with ID: adc88d5c-8950-4b71-8d50-98b14db99d63
Source Schema:LVSBX87
Target DB:
1. MD
2. CFG
3. STG
4. VAL
5. PROD
6. SBX
Enter your choice:2
2024-04-29 11:43:31 [main] INFO  com.alcon.mdlims.masterdata.service.DBConnectionService - Database connection successful with ID: 7cc30e3c-228b-4575-a973-0ad5761439f9
Target Schema:LVCFG87
Method type:METHOD
Folder Name:C19
User ID:SRA3
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Usecase: Prevalidate Import
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Source Connection:oracle.jdbc.driver.T4CConnection@62e20a76
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Source Schema: LVSBX87
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Target DB: oracle.jdbc.driver.T4CConnection@1d1f7216
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Target Schema: LVCFG87
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Method Type: METHOD
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - Folder Name: C19
2024-04-29 11:43:49 [main] INFO  com.alcon.mdlims.masterdata.bean.InputRequest - User ID: SRA3
2024-04-29 11:43:50 [main] INFO  com.alcon.mdlims.masterdata.dao.HierarchyRetrieval - Query for WORKITEM: SELECT TESTINGDEPARTMENTID, SECURITYDEPARTMENT, WORKAREADEPARTMENTID, EMBEDCHILDSAMPLEPLANID, EMBEDCHILDS
AMPLEPLANVERSIONID, APPLICABLESAMPLETYPEID, SECURITYUSER, AUTOASSIGNANALYSTID, QUANTITYUNITS, WORKITEMID, WORKITEMVERSIONID FROM LVSBX87.WORKITEM MDTB, (select  SFI.LINKSDCID, SFI.LINKKEYID1, SFI.LINKKEYID2, SFI
.LINKKEYID3 from lvsbx87.sysuserfolderitem SFI INNER JOIN lvsbx87.sysuserfolder SUF ON SUF.sysuserid = SFI.sysuserid and SUF.sysuserfolderid = SFI.sysuserfolderid WHERE SUF.folderlabel = 'C19' and SUF.sysuserid
= 'SRA3' ) FDT WHERE  MDTB.WORKITEMID = FDT.LINKKEYID1 AND  MDTB.WORKITEMVERSIONID = FDT.LINKKEYID2
2024-04-29 11:44:37 [main] INFO  com.alcon.mdlims.masterdata.dao.HierarchyRetrieval - Query for PARAMLIST: SELECT ACCREDITEDADDRESSID, ACCREDITEDADDRESSTYPE, APPROVALTYPEID, SECURITYDEPARTMENT, TESTINGDEPARTMENT
ID, WORKAREADEPARTMENTID, S_INSTRUMENTMODEL, S_INSTRUMENTTYPE, S_INSTRUMENTTYPE, LIMITRULEID, LIMITRULEVERSIONID, SECURITYUSER, AUTOASSIGNANALYSTID, PARAMLISTID, PARAMLISTVERSIONID, VARIANTID FROM LVSBX87.PARAML
IST MDTB, (select  SFI.LINKSDCID, SFI.LINKKEYID1, SFI.LINKKEYID2, SFI.LINKKEYID3 from lvsbx87.sysuserfolderitem SFI INNER JOIN lvsbx87.sysuserfolder SUF ON SUF.sysuserid = SFI.sysuserid and SUF.sysuserfolderid =
 SFI.sysuserfolderid WHERE SUF.folderlabel = 'C19' and SUF.sysuserid = 'SRA3' ) FDT WHERE  MDTB.PARAMLISTID = FDT.LINKKEYID1 AND  MDTB.PARAMLISTVERSIONID = FDT.LINKKEYID2 AND  MDTB.VARIANTID = FDT.LINKKEYID3
2024-04-29 11:45:50 [main] INFO  com.alcon.mdlims.masterdata.domain.TextFileWriter - Data written to file: PrevalidateImport_C19_SRA3_20240429_114349.txt
2024-04-29 11:45:50 [main] INFO  com.alcon.mdlims.masterdata.domain.TextFileWriter - Text File Writer closed successfully!
2024-04-29 11:45:50 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriverHelper - SourceDB closed successfully!
2024-04-29 11:45:51 [main] INFO  com.alcon.mdlims.masterdata.bootstrap.AppDriverHelper - TargetDB closed successfully!
```

Output:

```
PrevalidateImport_C19_SRA3_2024   ×   +
File   Edit   View

---------------------------------------------

Missing Keys in WORKITEM SDC:
Primary Keys:
WORKITEMID = PCR Test C19       WORKITEMVERSIONID = 1

Foreign Keys: SECURITYUSER = RAOBH1
WHERE Primary Keys: WORKITEMID = PCR Test C19 WORKITEMVERSIONID = 1


---------------------------------------------

---------------------------------------------

Missing Keys in WORKITEMITEM SDC:
Primary Keys:
WORKITEMID = PCR Test C19       WORKITEMVERSIONID = 1   WORKITEMITEMID = 1
WORKITEMID = PCR Test C19       WORKITEMVERSIONID = 1   WORKITEMITEMID = 2

Only Primary Keys missing
WHERE Primary Keys: WORKITEMID = PCR Test C19 WORKITEMVERSIONID = 1 WORKITEMITEMID = 1


Only Primary Keys missing
WHERE Primary Keys: WORKITEMID = PCR Test C19 WORKITEMVERSIONID = 1 WORKITEMITEMID = 2


---------------------------------------------

---------------------------------------------

Missing Keys in WORKITEMINSTRUMENT SDC:
No Primary Keys are missing
No Foreign Keys are missing
---------------------------------------------

---------------------------------------------

Missing Keys in WORKITEMREAGENTTYPE SDC:
Ln 1, Col 1     3,186 characters
```

CHAPTER 8
RECOMMENDATIONS

Flexible Training Schedule:

Offer interns the flexibility to choose their working hours, accommodating their individual preferences and schedules. By allowing interns to work during times that suit them best, you empower them to maintain a healthy work-life balance and optimize their productivity.

Structured Training Modules:

Develop structured training modules and resources to guide interns through the onboarding process and provide them with a clear roadmap for skill development. These modules should cover essential topics related to the company's technologies, processes, and culture, ensuring that interns receive comprehensive training that aligns with their learning objectives.

Mentorship Program:

Implement a mentorship program that pairs each intern with an experienced mentor who can provide guidance, feedback, and support throughout their internship. Mentors play a crucial role in facilitating learning, fostering professional growth, and helping interns navigate challenges effectively. Encourage mentors to schedule regular check-ins with their mentees and provide ongoing support and encouragement.

Feedback and Evaluation:

Provide interns with regular feedback and evaluation sessions to assess their progress, identify areas for improvement, and recognize their achievements. Feedback sessions should be constructive, specific, and actionable, highlighting interns' strengths and areas for development. Encourage interns to reflect on their experiences, set goals for their learning journey, and take ownership of their professional development. By fostering a culture of continuous feedback and improvement, you empower interns to maximize their potential and make meaningful contributions to the company.

CHAPTER 9
CONCLUSION

In summary, my internship at Alcon Laboratories (India) Pvt. Ltd. was a wonderful experience that provided me with valuable insights, skills, and lessons for my professional development.

Key Accomplishments:

- Successfully developed the Master Data Checker tool, contributing to improved data integrity and efficiency within the organization.
- Acquired technical skills in Java programming and SQL database management through hands-on project work.
- Participated in event with global leaders, gaining valuable insights and networking opportunities.

Challenges and Lessons Learned:

- Overcoming challenges through proactive networking, effective communication, and continuous learning.
- Recognizing the importance of project management skills and industry best practices for future success.
- Embracing opportunities for personal and professional growth, both technically and professionally.

I express my heartfelt gratitude to Alcon Laboratories for providing me with this invaluable internship opportunity. Special thanks to Tuhin Sajjad, Ajith Poulose, Vivek Kumar, Sujithra Murugan, Mayuri Kulkarni, Somya Singh, whose mentorship and support were instrumental in my growth and success during the internship. I am deeply appreciative of their guidance and encouragement throughout this journey.

CHAPTER 10
REFERENCES

- Developing Innovative Eye Care Treatments | Alcon.com
- Alcon - Wikipedia
- Alcon | LinkedIn